

1. If a mushroom is poisonous, then it is orange.

P = mushroom is poisonous, O = mushroom is orange

$$P \rightarrow O$$

2. If a mushroom is poisonous, then it has gills.

P = mushroom is poisonous, G = mushroom has gills

$$P \rightarrow G$$

3. If a mushroom is not poisonous, then it has pores.

P = mushroom is poisonous, H = mushroom has pores

$$\neg P \rightarrow H$$

4. If a mushroom is not poisonous, then it is found growing from the trunk of a living hardwood tree.

P = mushroom is poisonous, T = Growing from the trunk of a living hardwood tree

$$\neg P \rightarrow T$$

5. If a mushroom is bioluminescent, then it is poisonous.

B = mushroom is bioluminescent, P = mushroom is poisonous

$$B \rightarrow P$$

Mushroom A = $T \wedge G \wedge O$, To get all of these values as true (consistent):

$$(\neg P \rightarrow T) \wedge (P \rightarrow G) \wedge (P \rightarrow O) \equiv T$$

- If T, G, and O are true for mushroom A, then P can be either true or false ($\neg P$ being the complement to this value) for each of the above propositions to be true.
- This means that this mushroom could be either not poisonous or it could be poisonous, so it is not guaranteed to be safe to eat.

T/F T F/T T F/T T

$$(\neg P \rightarrow T) \wedge (P \rightarrow G) \wedge (P \rightarrow O) \equiv T$$

Mushroom B = $T \wedge H \wedge B \wedge (O \oplus \neg O)$, To get all these values as true (consistent):

$\neg O$ would be $\neg O \rightarrow \neg P$ following logical equivalences involving conditional statements.

$$(\neg P \rightarrow T) \wedge (\neg P \rightarrow H) \wedge (B \rightarrow P) \wedge ((P \rightarrow O) \oplus (\neg O \rightarrow \neg P)) \equiv T$$

- T, H, and B are set to true and looking at the $(B \rightarrow P)$ proposition, if B is set to true then it can only imply a true value for the proposition to be true.
- Since all propositions in this set must be true for mushroom B, then that means that for $(B \rightarrow P)$, P would have to be true. If P is true, then $\neg P$ would be false.
- For $((P \rightarrow O) \oplus (\neg O \rightarrow \neg P))$ to be true, only one of the two propositions within must be true. The mushroom could still be orange or brown and this proposition would evaluate to true.

- Because P is true that means that mushroom B is guaranteed to be poisonous and is therefore not safe to eat.

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$$(\neg P \rightarrow T) \wedge (\neg P \rightarrow H) \wedge (B \rightarrow P) \wedge ((P \rightarrow O) \oplus (\neg O \rightarrow \neg P)) \equiv T$$

Mushroom C = $T \wedge H \wedge \neg O \wedge \neg B$, To get all these values as true (consistent):

$\neg O$ would be $\neg O \rightarrow \neg P$ following logical equivalences involving conditional statements.

$\neg B$ would be $\neg P \rightarrow \neg B$ following logical equivalences involving conditional statements.

$$(\neg P \rightarrow T) \wedge (\neg P \rightarrow H) \wedge (\neg O \rightarrow \neg P) \wedge (\neg P \rightarrow \neg B) \equiv T$$

- T, H, $\neg O$, and $\neg B$ are all set to true. Because a true value must imply a true value for the proposition to evaluate to true, $(\neg O \rightarrow \neg P)$ would suggest that $\neg P$ would be true. As a result, all of the $\neg P$ would be set to true.
- Because $\neg P$ is true, this means that mushroom C is not poisonous and is therefore guaranteed to be safe to eat.

T T T T T T T T

$$(\neg P \rightarrow T) \wedge (\neg P \rightarrow H) \wedge (\neg O \rightarrow \neg P) \wedge (\neg P \rightarrow \neg B) \equiv T$$

Only **mushroom C** is guaranteed to be safe to eat.